A unique cubic-coupling cancellation locus for collinear constant-vorticity capillary–gravity triads

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Abstract

We classify a degeneracy of the leading resonant interaction in the co-propagating constant-vorticity capillary–gravity wave model of Ivanov and Martin. Let $g, \sigma > 0$, let $k_1, k_2 > 0$ be distinct, and set $k_3 = k_1 + k_2$, where σ is surface tension divided by density. Starting from the polynomial dispersion relations and $\omega_3 = \omega_1 + \omega_2$, we obtain all real solutions without squaring a radical equation. There are exactly two: a common-positive frequency solution and its simultaneous frequency–vorticity sign reversal. On either sheet the cubic interaction numerator is a rational function of $q_j = k_j(g + \sigma k_j^2)$. Requiring this numerator to vanish reduces exactly to a quartic equation for $r = g/\sigma$. Its four linear factors are positive and increasing on $r \geq 0$, while its value at the origin is strictly negative. Hence every distinct positive pair (k_1, k_2) has exactly one $r_* > 0$ at which the linear resonance persists but the retained cubic triad coefficient vanishes. The vorticity is negative on the positive-frequency sheet and positive on its sign-reversed companion. The result concerns only the source model's short-wave, cubic-order envelope truncation; it neither suppresses higher-order transfer nor establishes a finite-depth or full-Euler cancellation.

Keywords: capillary–gravity waves; constant vorticity; resonant triad; Hamiltonian interaction coefficient; exact classification

Mathematics Subject Classification (2020): 76B15, 76B45, 35Q35

1 Introduction

Three-wave resonance requires both a wave-vector relation and a frequency relation. Constant vorticity can enable such resonances for collinear capillary and capillary–gravity waves even when the corresponding irrotational configuration is nonresonant [1, 2]. Ivanov and Martin [3] derived a weakly nonlinear Hamiltonian and envelope system for three distinct, co-propagating modes in a constant-vorticity flow. The resulting dynamics contains a common cubic interaction numerator, so linear resonance and nonlinear coupling are logically separate conditions.

The linear-resonance part is substantially understood. Martin [2] derived the vorticity magnitude with an explicit sign restoration, and Martin and Basu [4] obtained the equation-equivalent positive-frequency root and its uniqueness in the capillary regime for a surface layer of piecewise-constant vorticity. Accordingly, neither that magnitude nor negative-vorticity uniqueness is claimed here as new. For the pure-capillary boundary $g = 0$, Constantin, Kartashova, and Wahlén [5] gave a Hamiltonian coupling coefficient and studied resonance clusters.

Our question is instead simultaneous: when do the dispersion equations, linear resonance, and the cubic coefficient from [3] vanish together? We first retain the unrestricted real polynomial dispersion system and show that it has exactly two total-sign-related sheets. We then reduce the

simultaneous zero to one strictly increasing quartic in $r = g/\sigma$, proving existence and uniqueness of a positive zero for every distinct positive wave-number pair. This is not the same mechanism as the fixed-amplitude and fixed-phase no-exchange states studied by Chabane and Choi [6].

The scope is deliberately narrow. The calculation uses the short-wave Fourier-multiplier replacement and the cubic, slow-envelope truncation of [3]. A zero of its cubic coefficient is not a statement about all perturbative orders, finite depth, dissipation, or an experiment.

2 Model and conventions

The parent setting is a two-dimensional inviscid, incompressible fluid with a free surface, a flat bed, gravity $g > 0$, and constant vorticity $\gamma = u_y - v_x$. We write

$$\sigma = \frac{\widetilde{\sigma}}{\rho} > 0,$$

where $\widetilde{\sigma}$ is surface tension and ρ is density. Thus

$$[k] = L^{-1}, \qquad [\omega] = [\gamma] = T^{-1}, \qquad [g] = LT^{-2}, \qquad [\sigma] = L^3T^{-2}.$$

At the short-wave level retained in [3], a positive co-propagating wave number satisfies

$$\omega^2 + \gamma\omega - k(g + \sigma k^2) = 0. \tag{1}$$

The selected carrier frequency is the positive root

$$\omega_\gamma^+(k) = \frac{-\gamma + \sqrt{\gamma^2 + 4k(g + \sigma k^2)}}{2} > 0. \tag{2}$$

Let $k_1, k_2 > 0$, $k_1 \neq k_2$, and $k_3 = k_1 + k_2$. The resonance condition is

$$\omega_3 = \omega_1 + \omega_2. \tag{3}$$

For convenience define

$$q_j = k_j(g + \sigma k_j^2), \qquad C = \gamma^2 + 2\gamma\omega_3 + 2\omega_1\omega_2. \tag{4}$$

Direct summation of the four resonant cubic Hamiltonian terms in [3, Eq. (4.31)] gives C. Each nonlinear right-hand side of the three envelope equations in its Eq. (4.32) is proportional to $k_jC/(2\omega_j + \gamma)$, multiplied by the appropriate quadratic amplitude monomial. Hence $C = 0$, provided the denominators are regular, removes precisely the displayed cubic resonant interaction.

3 Complete real resonance sheets

Write

$$\begin{gathered} a = q_1, \qquad b = q_2, \qquad c = q_3, \qquad \ell = c - a - b = 3\sigma k_1k_2k_3, \\ p = \frac{\ell}{2}, \qquad Q = c^2 - (a - b)^2, \qquad N = \ell^2 - 4ab, \qquad D = 2\ell Q. \end{gathered} \tag{5}$$

Because $q(k) = k(g + \sigma k^2)$ is strictly increasing for $k > 0$, $c > a, b$, and therefore

$$Q = (c - a + b)(c + a - b) > 0, \qquad D > 0.$$

Theorem 1 (Two-sheet classification)**.** *Every real solution of the three dispersion polynomials* $\omega_j^2 + \gamma\omega_j - q_j = 0$ *and* (3) *is, for one* $\varepsilon \in \{+1, -1\}$*,*

$$\begin{aligned} \omega_1^{(\varepsilon)} &= \varepsilon\sqrt{p}\,\frac{c+a-b}{\sqrt{Q}}, \quad \omega_2^{(\varepsilon)} = \varepsilon\sqrt{p}\,\frac{c-a+b}{\sqrt{Q}}, \\ \omega_3^{(\varepsilon)} &= \varepsilon\,\frac{2c\sqrt{p}}{\sqrt{Q}}, \qquad\qquad \gamma^{(\varepsilon)} = -\varepsilon\,\frac{N}{\sqrt{D}}. \end{aligned} \tag{6}$$

In particular, the physical positive-frequency sheet is $\varepsilon = +1$*. The other solution is the simultaneous sign reversal* $(\omega_j, \gamma) \mapsto (-\omega_j, -\gamma)$*; there are no mixed-sign real branches.*

Proof. Subtract the first two dispersion equations from the third after using $\omega_3 = \omega_1 + \omega_2$. This gives

$$2\omega_1\omega_2 = \ell > 0. \tag{7}$$

Thus ω_1 and ω_2 are nonzero and have the same sign, which is also the sign of ω_3. Differences of the dispersion equations also yield

$$c + a - b = 2\omega_1(\omega_3 + \gamma), \qquad c - a + b = 2\omega_2(\omega_3 + \gamma). \tag{8}$$

The positive left-hand sides show that the common factor does not vanish. Taking the ratio in (8) and combining it with (7) fixes ω_1, ω_2, up to their common sign, as the first line of (6). Their sum gives the displayed ω_3. Finally, substituting either frequency into its dispersion equation and simplifying gives $\gamma = -\varepsilon N/\sqrt{D}$. Direct substitution of (6) verifies all four equations. The product, common sign, and ratio arguments force every real solution to this list, proving completeness. □

Remark 1. Eliminating the frequencies gives only

$$D\gamma^2 - N^2 = 0,$$

which loses the sheet label. At $N = 0$, both solutions have $\gamma = 0$, but their three nonzero frequencies still have opposite signs.

For example, with $g = \sigma = 1$ and $(k_1, k_2, k_3) = (1, 3, 4)$, the two solutions are

$$(\gamma, \omega_1, \omega_2, \omega_3) = \frac{(-11, 15, 36, 51)}{\sqrt{30}}, \qquad \frac{(11, -15, -36, -51)}{\sqrt{30}}. \tag{9}$$

The first is the selected positive-frequency solution.

4 The unique cubic-coupling zero

Lemma 1. *On either resonance sheet in Theorem 1,*

$$C = \frac{4ab}{\ell} - \gamma^2 = \frac{16ab(c-a)(c-b) - \ell^4}{2\ell\{c^2 - (a-b)^2\}}. \tag{10}$$

In particular, C *is invariant under total sign reversal.*

Proof. The dispersion equations and (7) give

$$ab = \omega_1\omega_2(\omega_1 + \gamma)(\omega_2 + \gamma) = \frac{\ell}{2}\left(\frac{\ell}{2} + \gamma\omega_3 + \gamma^2\right).$$

Comparing this identity with (4) proves the first equality in (10). Inserting $\gamma^2 = N^2/D$ and factoring proves the second. □

Set $x = k_1$, $y = k_2$, and

$$r = \frac{g}{\sigma},$$

which has dimensions L^{-2}.

Theorem 2 (Unique cancellation locus)**.** *For every distinct $x, y > 0$, there is exactly one $r_*(x, y) > 0$ for which the dispersion equations, resonance, and $C = 0$ hold simultaneously. It is the unique positive root of*

$$\begin{aligned} P_{x,y}(r) = 16(r + x^2)(r + y^2)(r + 3x^2 + 3xy + y^2) \\ \times (r + x^2 + 3xy + 3y^2) - 81x^2y^2(x + y)^4. \end{aligned} \tag{11}$$

Moreover, $C < 0$ for $0 \le r < r_*$ and $C > 0$ for $r > r_*$ on both sheets. At r_*, the positive-frequency sheet has $\gamma < 0$, while its total-sign companion has $\gamma > 0$.

Proof. The numerator of (10) factors exactly as

$$\sigma^4 x^2 y^2 P_{x,y}(r).$$

The removed factor and the denominator in (10) are strictly positive. Each of the four linear factors in the product in (11) is positive and strictly increasing on $r \ge 0$; therefore $P_{x,y}$ is strictly increasing there. At the endpoint,

$$P_{x,y}(0) = -x^2y^2\{33(x + y)^4 - 16x^2y^2\} < 0, \tag{12}$$

and $P_{x,y}(r) \to +\infty$ as $r \to +\infty$. The intermediate value theorem and strict monotonicity prove existence, uniqueness, and the sign change.

On the positive-frequency sheet, solving $C = 0$ at the simultaneous solution gives

$$\gamma = -(\omega_1 + \omega_2) \pm \sqrt{\omega_1^2 + \omega_2^2}.$$

Both values are negative because $\sqrt{\omega_1^2 + \omega_2^2} < \omega_1 + \omega_2$. The other sheet is the total sign reversal, so its vorticity is positive. □

The amplitude denominators are nonsingular at the zero:

$$2\omega_j + \gamma = \sqrt{\gamma^2 + 4q_j} > 0 \quad \text{on the positive-frequency sheet.} \tag{13}$$

For $(k_1, k_2) = (1, 3)$,

$$r_* = 0.5179682428904992165\ldots, \qquad \gamma_* = -2.194536665566565812\ldots$$

when $\sigma = 1$. These decimals illustrate the exact result; they are not used in its proof.

5 Relation to earlier formulas

The convention in Martin [2] uses

$$\omega = \frac{\gamma_M + \sqrt{\gamma_M^2 + 4q}}{2},$$

so its scalar equation maps to the present one by $\gamma_M = -\gamma$. Its squared vorticity formula agrees with $\gamma^2 = N^2/D$, while its unsquared equation restores the sign. Likewise, after the surface-current Doppler terms cancel under $k_1 + k_2 = k_3$, the λ_1 dispersion branch of Martin and Basu [4] is (2). Their variables reduce to

$$E = -\ell/\sigma, \qquad F = N/\sigma^2, \qquad 4\sigma\Sigma^2 = N^2/D.$$

Their result therefore supplies prior linear-resonance input, but it does not contain C or the simultaneous quartic (11).

The pure-capillary Hamiltonian study [5] is the closest coupling-coefficient antecedent. It lies at $r = 0$, where (12) shows that the present coefficient is not zero; the cancellation locus is strictly inside $r > 0$.

Two formula issues should be kept version-specific. In the inspected arXiv-v1 rendering of [3], the displayed Eq. (3.3) lacks the square root needed when recovering a vorticity from the squared resonance relation. It is dimensionally inconsistent as printed. For $(k_1, k_2, k_3) = (1, 3, 4)$ and $g = \sigma = 1$, its literal value is $11/5760$, for which direct evaluation of the selected branch gives the nonzero resonance residual

$$\omega_1 + \omega_2 - \omega_3 = -1.3557266246176232\ldots.$$

Equation (9) gives the substitution-consistent replacement. The sentence after its Eq. (4.33) also prints $\gamma = \omega_1 + \omega_2 \pm \sqrt{\omega_1^2 + \omega_2^2}$. For positive frequencies these values are positive and cannot zero the all-positive-term expression (4). The leading sign must instead be negative, as in the proof of Theorem 2, and dispersion and resonance must still be imposed simultaneously.

These observations concern the checked arXiv v1 only. We did not inspect the equation rendering in the version of record, and we do not assert the absence of an unindexed author correction or a formal publisher correction.

6 Physical interpretation and limitations

At r_*, the linear phase matching (3) remains exact, while the common numerator of all three displayed cubic envelope couplings vanishes. Thus the leading retained resonant exchange degenerates. Calling this *no energy exchange* without an order qualifier would be too strong: quartic and higher Hamiltonian terms, finite-depth corrections, dissipation, or other modes may supply the first nonzero transfer. The rescaling used to normalize a nonzero three-wave system is singular at $C = 0$ and cannot be used there.

The analysis does not test the replacement of the finite-depth multiplier by its short-wave sign limit. It excludes counter-propagating and noncollinear triads, and it does not address stability, detuning width, experimental accessibility, or the full Euler evolution. The all-negative-frequency solution is algebraic bookkeeping for the unrestricted polynomial system, not a second physical carrier at the same fixed vorticity.

7 Conclusion

The source-model resonance manifold has exactly two total-sign-related real sheets. On that manifold, the cubic interaction coefficient admits the closed form (10), and its zero is equivalent to the monotone quartic (11). Every distinct positive wave-number pair therefore determines one and only one positive gravity-to-capillarity ratio at which linear resonance survives but the retained cubic coupling vanishes. This exact separation of phase matching from leading nonlinear transfer is the principal result; determining the first nonzero omitted interaction is a separate problem.

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